

Hasan Shahriar

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APPOINTMENT

Assistant Professor
UNIVERSITY OF ARKANSAS
Department of Electrical Engineering and Computer Science

Fall 2026–Present
Fayetteville, Arkansas, USA

EDUCATION

PhD in Computer Science
VIRGINIA TECH

Jan 2021–May 2026
Arlington, Virginia, USA

- Dissertation: *Toward Trustworthy Autonomous Cyber-Physical Systems: Robust Machine Learning for Secure Sensing, Perception, and Control*
- Advisor: Dr. Wenjing Lou

MS in Computer Engineering
FLORIDA INTERNATIONAL UNIVERSITY

Jan 2019–Dec 2020
Miami, Florida, USA

- Thesis: *Deception Defense against Stealthy Attacks in Power Grids*
- Advisor: Dr. Mohammad Ashiqur Rahman

BSc in Electrical and Electronic Engineering
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Feb 2011–Mar 2016
Dhaka, Bangladesh

- Thesis: *Transient Stability Analysis of Smart Grids with Impacts of Distributed Generation*
- Advisor: Dr. Md Forkan Uddin

RESEARCH INTERESTS

My research spans cyber-physical systems (CPS), artificial intelligence (AI), and cybersecurity, with a focus on enabling secure and resilient autonomous systems. I develop cross-layer defense frameworks to protect the physical layer (*intrusion detection and cyber-resilient CPS*), the intelligence layer (*trustworthy and resilient AI*), and the *cross-layer ML infrastructure* that integrates them, ensuring reliable operation even in the presence of coordinated cyber-physical and AI-driven threats.

AWARDS, FELLOWSHIPS, & GRANTS

- [A12] NSF Cyber SMART Distinguished Dissertation Award, Virginia Tech site, 2026.
- [A11] Amazon Fellowship, Amazon-VT Initiative for Efficient and Robust Machine Learning, 2024-2025.
- [A10] Student Travel Grant, Maritime Cybersecurity Hackathon, 2025.
- [A9] Student Travel Grant for Attending IEEE ICDCS, U.S. National Science Foundation, 2024.
- [A8] Student Travel Grant, CyberTruck Challenge, 2024.
- [A7] Best Paper Runner Up Award, Symposium on Vehicle Security and Privacy (VehicleSec), 2023.
- [A6] Student Travel Grant, Inaugural Symposium on Vehicle Security and Privacy (VehicleSec), 2023.
- [A5] Fellowship for Graduate Student First-Author Papers, Graduate School, Virginia Tech, 2023.
- [A4] Bangladesh-Sweden Trust Fund Scholarship, July 2021.
- [A3] Student Travel Grant for Attending ACM WiSec, U.S. Army Research Office, 2019.

[A2] **Admission Test Excellency Scholarship**, *Bangladesh University of Engineering and Technology*, 2011.

[A1] **Education Board Scholarship**, *Government of Bangladesh*, 2008 & 2010.

PUBLICATIONS

Journal Articles

- [J3] **Md Hasan Shahriar**, Mohammad Raashid Ansari, M. S. Haque, Jean-Philippe Monteuiis, Cong Chen, Jonathan Petit, Y. Thomas Hou, Wenjing Lou. “VehiGAN: Generative Adversarial Networks for Adversarially Robust V2X Misbehavior Detection Systems.” In *ACM Transactions on Cyber-Physical Systems (ACM TCPS)*, 2025. (Impact Factor: 2.0).
- [J2] **Md Hasan Shahriar**, Y. Xiao, P. Moriano, Wenjing Lou, Y. Thomas Hou. “CANShield: Deep Learning-Based Intrusion Detection Framework for Controller Area Networks at the Signal-Level.” In *IEEE Internet of Things Journal (IEEE IoT-J)*, 2023. (Impact Factor: 10.6).
- [J1] **Md Hasan Shahriar**, M. A. Rahman, M. Jafari, S. Paudyal. “Formal Analytics for Stealthy Attacks against Contingency Analysis in Power Grids.” In *Sustainable Energy, Grids and Networks (SEGAN)*, 2024. (Impact Factor: 5.6).

Conference Papers (Selected)

- [C17] Hexuan Yu, Md Mohaimin Barat, Shaoyu Li, **Md Hasan Shahriar**, Yang Xiao, Panos Papadimitratos, Y. Thomas Hou, Wenjing Lou. “V-PASS: Sybil-Resistant Pseudonym Self-Provisioning for V2X.” Accepted to appear in *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2026.
- [C16] **Md Hasan Shahriar**, Ning Wang, Amit Kumar Sikder, Naren Ramakrishnan, Y. Thomas Hou, Wenjing Lou. “Noise, Why Can’t You Bend? Detecting Adversarial Perturbations in Wireless Sensing via Structural Fragility” Accepted to appear in *ACM ASIA Conference on Computer and Communications Security (AsiaCCS)*, 2026.
- [C15] **Md Hasan Shahriar**, Mohaimin Al Barat, Harshavardhan Sundar, Ning Zhang, Naren Ramakrishnan, Y. Thomas Hou, Wenjing Lou. “Detecting Temporal Misalignment Attacks in Multimodal Fusion for Autonomous Driving” In *Proceedings of The International Conference on Learning Representations (ICLR)*, 2026. (Acceptance rate: 28%).
- [C14] **Md Hasan Shahriar**, Mohaimin Al Barat, Harshavardhan Sundar, Ning Zhang, Naren Ramakrishnan, Y. Thomas Hou, Wenjing Lou. “Temporal Misalignment Attacks against Multimodal Perception in Autonomous Driving” In *Proceedings of IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2026. (Acceptance rate: 26%).
- [C13] Kuan Yu Chen, **Md Hasan Shahriar**, Wen Wei Li, Shi Cho Cha, Wenjing Lou. “HOTWIRE: Real-World Impersonation and Discharge Attacks on Electric Vehicle Charging Systems” Accepted to appear in *USENIX WOOT Conference on Offensive Technologies (WOOT)*, 2026. (Acceptance rate: 30%).
- [C12] **Md Hasan Shahriar**, Ning Wang, Naren Ramakrishnan, Y. Thomas Hou, Wenjing Lou. “Let the Noise Speak: Harnessing Noise for a Unified Defense Against Adversarial and Backdoor Attacks.” In *Proceedings of European Symposium on Research in Computer Security (ESORICS)*, 2025. (Acceptance rate: 17%).
- [C11] **Md Hasan Shahriar**, Mohammad Raashid Ansari, Jean-Philippe Monteuiis, Cong Chen, Jonathan Petit, Y. Thomas Hou, Wenjing Lou. “VehiGAN: Generative Adversarial Networks for Adversarially Robust V2X Misbehavior Detection Systems.” In *Proceedings of IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2024. (Acceptance rate: 21%).
- [C10] **Md Hasan Shahriar**, Wenjing Lou, Y. Thomas Hou. “CANtropy: Time Series Feature Extraction-Based Intrusion Detection Systems for Controller Area Networks.” In *Proceedings of Symposium on Vehicle Security and Privacy (VehicleSec)*, 2023. **Best Paper Runner-Up Award**. (Acceptance rate: 36.0%).
- [C9] S. Shi, Y. Xiao, C. Du, **Md Hasan Shahriar**, A. Li, Ning Zhang, Y. Thomas Hou, Wenjing Lou. “MS-PTP: Protecting Network Timing from Byzantine Attacks.” In *Proceedings of ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2023. (Acceptance rate: 25.4%).
- [C8] **Md Hasan Shahriar**, Y. Xiao, P. Moriano, Wenjing Lou, Y. Thomas Hou. “CANShield: Signal-based Intrusion Detection for Controller Area Networks.” In *Embedded Security in Cars (ESCAR)*, 2022.
- [C7] **Md Hasan Shahriar**, Mohammad Ashiqur Rahman, Nur Imtiazul Haque, Badrul Chowdhury, S. G. Whisenant. “iDDAF: An Intelligent Deceptive Data Acquisition Framework for Secure Cyber-Physical Systems.” In *Proceed-*

ings of *EAI International Conference on Security and Privacy in Communication Networks (SecureComm)*, 2021. (Acceptance rate: 34%)

- [C6] **Md Hasan Shahriar**, Mohammad Ashiqur Rahman, Nur Imtiazul Haque, Badrul Chowdhury. “DDAF: Deceptive Data Acquisition Framework against Stealthy Attacks in Cyber-Physical Systems.” In *Proceedings of IEEE 45th International Conference on Software Engineering (COMPSAC)*, 2021. (Acceptance rate: 27%)
- [C5] **Md Hasan Shahriar**, Alvi Ataur Khalil, Mohammad Ashiqur Rahman, Mohammad Hossein Manshaei, Dong Chen. “iAttackGen: Generative Synthesis of False Data Injection Attacks in Cyber-Physical Systems.” In *Proceedings of IEEE Conference on Communications and Network Security (CNS)*, 2021. (Acceptance rate: 26%)
- [C4] M. Jafari, **Md Hasan Shahriar**, Mohammad Ashiqur Rahman, S. Paudyal. “False Relay Operation Attacks in Power Systems with High Renewables.” In *Proceedings of IEEE Power & Energy Society General Meeting (PESGM)*, 2021.
- [C3] Nur Imtiazul Haque, **Md Hasan Shahriar**, Md Golam Dastgir, Anjan Debnath, Imtiaz Parvez, Arif Sarwat, and Mohammad Ashiqur Rahman. “Machine Learning in Generation, Detection, and Mitigation of Cyberattacks in Smart Grid: A Survey.” In *Proceedings of North American Power Symposium (NAPS)*, 2021.
- [C2] **Md Hasan Shahriar**, Nur Imtiazul Haque, Mohammad Ashiqur Rahman, Miguel Alonso Jr. “G-IDS: Generative Adversarial Networks Assisted Intrusion Detection System.” In *Proceedings of IEEE 45th International Conference on Software Engineering (COMPSAC)*, 2020. (Acceptance rate: 24%)
- [C1] **Md Hasan Shahriar**, Md Jawwad Sadiq, and Md Forkan Uddin. “Stability Analysis of Grid-connected PV Array Under Maximum Power Point Tracking”. In *Proceedings of International Conference on Electrical and Computer Engineering (ICECE)*, 2016.

Theses

- [T3] **Md Hasan Shahriar**. “Toward Trustworthy Cyber-physical Systems: Robust Machine Learning for Secure Sensing, Perception, and Control” In *Virginia Tech Theses and Dissertations*, 2026 (Anticipated).
- [T2] **Md Hasan Shahriar**. “Deception Defense against Stealthy Attacks in Power Grids.” In *Florida International University Theses and Dissertations*, 2020.
- [T1] **Md Hasan Shahriar**. “Transient Stability Analysis of Smart Grids with Impacts of Distributed Generation.” In *Bangladesh University of Engineering and Technology Theses and Dissertations*, 2016.

PRESENTATIONS & TALKS

Invited Research Talks:

- “Temporal Misalignment Attack against Multimodal Fusion in Autonomous Driving”
4th Workshop on Future Automotive Research Datasets, November 2025.
- “Security of Connected and Autonomous Vehicles: From In-vehicular Networks to Multimodal Fusion”
Amazon VT Initiative Kickoff, Invited Talk, Blacksburg, VA, Fall 2024.
- “Generating State-of-the-art V2X Misbehavior Detection Dataset and a Robust Detection Approach”
3rd Workshop on Future Automotive Research Datasets, April 2024.
- “CANShield: Signal-based Intrusion Detection for Controller Area Networks”,
1st Workshop on Future Automotive Research Datasets, April 2021 and
ACIC-DoD ROLLAGE TEM, November 2021.
- “A Survey on CAN Intrusion Detection Dataset”
1st Workshop on Automotive Research Datasets, November 2021.
- “Deception-based Defense against False Data Injection Attacks in Power Grids”
CAPER Meeting (Virtual), Fall 2020.

Paper Presentations:

- **IEEE SaTML 2026**, “Temporal Misalignment Attack against Multimodal Fusion in Autonomous Driving.”, April 2026.
- **ESORICS 2025**, “Let the Noise Speak: Harnessing Noise for a Unified Defense Against Adversarial and Backdoor Attacks.”, September 2025.

- **IEEE ICDCS 2024**, “VehiGAN: Generative Adversarial Networks for Adversarially Robust V2X Misbehavior Detection Systems”, July 2024.
- **VehicleSec 2023**, “CANtropy: Time Series Feature Extraction-Based Intrusion Detection Systems for Controller Area Networks”, February 2023.
- **ESCAR USA 2022**, “CANShield: Signal-based Intrusion Detection for Controller Area Networks”, June 2022.
- **IEEE COMPSAC 2021**, “DDAF: Deceptive Data Acquisition Framework against Stealthy Attacks in Cyber-Physical Systems”, July 2021.
- **EAI SecureComm 2021**, “iDDAF: An Intelligent Deceptive Data Acquisition Framework for Secure Cyber-physical Systems”, September 2021.

Poster Presentations:

- **ICLR’26**, “Detecting Temporal Misalignment Attacks in Multimodal Fusion for Autonomous Driving,” M. H. Shahriar, M. A. Barat, H. Sundar, N. Zhang, N. Ramakrishnan, Y. T. Hou, W. Lou. International Conference on Learning Representations (ICLR), 2026.
- **SaTML’26**, “Temporal Misalignment Attacks against Multimodal Perception in Autonomous Driving,” M. H. Shahriar, M. A. Barat, H. Sundar, N. Zhang, N. Ramakrishnan, Y. T. Hou, W. Lou. IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), 2026.
- **Amazon-VT’24**, “VehiGAN: Generative Adversarial Networks for Adversarially Robust V2X Misbehavior Detection Systems,” M. H. Shahriar, M. R. Ansari, J.-P. Monteuiis, C. Chen, J. Petit, Y. T. Hou, W. Lou, Fall Kickoff Meeting of Amazon-VT Initiative, Blacksburg, VA, 2024.
- **VehicleSec’23**, “CANtropy: Time Series Feature Extraction-Based Intrusion Detection Systems for Controller Area Networks,” M. H. Shahriar, W. Lou, Y. T. Hou. Symposium on Vehicle Security and Privacy (VehicleSec), 2023.
- **WiSec’19 & FICS’19**, “Poster: False Data Injection Attacks against Contingency Analysis in Power Grids,” M. Rahman, M.H. Shahriar, R. Masum, ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec), 2019 & also in FICS Research Annual Conference on Cybersecurity, University of Florida, 2019.

RESEARCH APPOINTMENTS

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| <p>[R3] Graduate Student Researcher
 COMPLEX NETWORK AND SECURITY RESEARCH (CNSR) LAB, VIRGINIA TECH
 Advisor: Dr. Wenjing Lou
 My Ph.D. research integrates cybersecurity, machine learning, and CPS to develop robust, resilient, and trustworthy AI for connected and autonomous vehicles. I designed intrusion detection systems for CAN and V2X networks to counter stealthy and adversarial attacks [C8,C10,C11,J2,J3], investigated network-induced and multimodal fusion attacks on autonomous perception [C14,C15], and developed a unified defense framework that mitigates both adversarial and backdoor ML threats across diverse modalities [C12,C16]. In addition, I mentored several graduate students whose work contributed to peer-reviewed and ongoing publications.</p> | <p>2021 – 2026
 Arlington, Virginia, USA</p> |
| <p>[R2] Graduate Student Researcher
 ANALYTICS FOR CYBER DEFENSE (ACyD) LAB, FLORIDA INTERNATIONAL UNIVERSITY
 Advisor: Dr. Mohammad Ashiqur Rahman
 My M.S. research focused on securing cyber-physical systems, particularly detecting and mitigating stealthy threats in the smart grid. I developed a threat synthesizer combining formal methods and GANs to model complex attack behaviors [J1, C4, C5]. Building on this, I proposed a deception-based moving target defense to counter stealthy intrusions [C6, C7] and explored GAN-based defense models for network security [C2], while mentoring several undergraduate researchers.</p> | <p>2019–2020
 Miami, Florida, USA</p> |
| <p>[R1] Undergraduate Student Researcher
 BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
 Advisor: Dr. Forkan Uddin
 My undergraduate research focused on analyzing the robustness of different Maximum Power Point Tracking algorithms for the smart grid under cyberattacks and developing strategies to improve system resilience [C1].</p> | <p>2014–2016
 Dhaka, Bangladesh</p> |

INDUSTRIAL EXPERIENCE

- [I3] **Interim Engineering Intern** **May 2022–Aug 2022**
San Diego, California, USA
QUALCOMM INCORPORATED
Manager: *Jonathan Petit*, **Mentor:** *Jean-Philippe Monteuis*
 - Executed real-world adversarial attacks on traffic sign detection systems to quantify adversarial robustness.
 - Investigated the transferability of adversarial examples across diverse object detection models.
 - Developed DVC-based pipelines for systematic dataset management and reproducible experiments.
- [I2] **Interim Engineering Intern** **May 2022–Aug 2022**
Boxborough, Massachusetts, USA
QUALCOMM INCORPORATED
Manager: *Jonathan Petit*, **Mentor:** *Rashed Ansari*
 - Researched and evaluated generative AI models (GANs) to synthesize realistic yet fake V2X messages.
 - Designed a GAN-based misbehavior detection to effectively detect anomalous basic safety messages (BSMs).
 - Continued the collaboration beyond internship and extended this project, which resulted in [C11,J3].
- [I1] **Assistant Engineer (Electrical)** **Sep 2017 – Dec 2018**
Dhaka, Bangladesh
ELECTRICITY GENERATION COMPANY BANGLADESH LTD.
 - Operated a 2x120 MW gas turbine power plant by coordinating with the national load dispatch center.
 - Developed operational and maintenance schedules to minimize downtime through proactive planning.

TEACHING EXPERIENCE

- Guest Lecturer, Department of Computer Science** **Spring 2026**
Blacksburg, VA
VIRGINIA TECH
Guest lectured for the following graduate course:
 - **CS 5984: Fundamentals of Information Security** Spring 2026
- Lecturer, Department of Computer Science** **May 2016 – May 2017**
Uttara, Dhaka, Bangladesh
UTTARA UNIVERSITY
Taught the following undergraduate courses and led corresponding lab sessions:
 - **EE 101: Electrical Circuits** Fall 2016
 - **EE 205: Basic Electronics** Spring 2017
 - **EE 210: Digital Logic Design** Fall 2016, Spring 2017
 - **EE 315: Microprocessor Interfacing and Low-Level Language** Fall 2016, Spring 2017

STUDENT MENTORSHIP

- **Kuan Yu Chen (MS, National Taiwan University of Science and Technology):** EV Charging Security.
- **Md Mohaimin Al Barat (PhD, Virginia Tech):** Security of Multimodal Fusion in Autonomous Driving.
- **Sydney Johns (PhD, Virginia Tech):** Practical Intrusion Detection Systems for Cyber-physical Systems.
- **Md Shahedul Haque (MS, Virginia Tech):** Defacing Technique in MRI data for Privacy Preserving ML.
- **Samara Ruiz Sandoval (Undergrad, Florida International University):** ML for Security of Smart Grid.

FEATURED

NSF Cyber SMART Center announces inaugural recipients of its Distinguished Dissertation Award, Cyber SMART News, 06/30/2026

Amazon-Virginia Tech Initiative awards two student fellowships, five faculty research awards, VT News, 10/22/2024

PROFESSIONAL SERVICES

Technical Program Committee:

- IEEE Int. Conf. on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm) 2026
- IEEE Conference on Communications and Network Security (IEEE CNS) 2026

Artifact Evaluation Committee:

- ACM Conference on Computer and Communications Security (CCS) 2025, 2026
- USENIX WOOT Conference on Offensive Technologies (WOOT) 2026

Journal Reviewer:

- IEEE Transactions on Vehicular Technology (TVT) 2024–2026
- IEEE Transactions on Dependable and Secure Computing (TDSC) 2026
- IEEE Internet of Things Journal (IoT-J) 2025
- IEEE Transactions on Big Data (TBD) 2025
- ACM Transactions on Cyber-Physical Systems (TCPS) 2025
- IEEE Transactions on Computers (TC) 2024
- Computers & Security (C&S) 2024
- IEEE Sensors Journal (SJ) 2023
- IEEE Transactions on Information Forensics and Security (TIFS) 2023
- Vehicular Communications (VehiCom) 2023
- IEEE Power & Energy Society Transactions on Power Systems (IEEE PES) 2021
- International Journal of Electronic Security and Digital Forensics (IJESDF) 2021

External Conference Reviewer:

- ACM ASIA Conference on Computer and Communications Security (AsiaCCS) 2025
- IEEE Symposium on Security and Privacy (IEEE S&P) 2022–2025
- European Symposium on Research in Computer Security (ESORICS) 2022–2024
- ACM Conference on Security and Privacy in Wireless and Mobile Networks (ACM WiSec) 2022–2025
- IEEE Conference on Communications and Network Security (IEEE CNS) 2022–2024
- International Conference on Computer Communication and Networks (ICCCN) 2023
- IEEE International Conference on Distributed Computing Systems (ICDCS) 2022
- IEEE International Conference on Communications (ICC) 2020
- International Symposium on Network Systems Security (NSysS) 2019

Community and Outreach Involvement:

- Student Volunteer, NSF CISE CAREER Workshop, 2026
- Student Volunteer, IEEE International Conference on Distributed Computing Systems (ICDCS), 2024
- Campus Representative, Graduate Student Assembly (GSA) – DC Region, Virginia Tech, 2023
- Session Chair, IEOM North American Industrial Engineering and Operations Management Conference, 2022
- Judge, Engineering Section, Northern Virginia Regional Science Fair, 2022
- Student Volunteer, ACM Conference on Security & Privacy in Wireless and Mobile Networks (WiSec), 2020

REFERENCES

Available upon request.